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(54) **AMORPHOUS METAL RIBBON, METHOD FOR MANUFACTURING AMORPHOUS METAL RIBBON, AND MAGNETIC CORE**

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(51) **Int. Cl.**

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B23K 101/36	(2006.01)
H01F 41/02	(2006.01)

(52) **U.S. Cl.**

CPC ***H01F 3/04*** (2013.01); ***B23K 26/364*** (2015.10); ***H01F 41/02*** (2013.01); *B23K 2101/36* (2018.08)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

An amorphous metal ribbon includes a plurality of laser irradiation mark rows each including a plurality of laser irradiation marks arranged in a row, in which when a distance between the laser irradiation mark rows that are adjacent to each other is set as d**1**, a distance between the laser irradiation marks in the laser irradiation mark row is set as d**2**, a diameter of the laser irradiation mark is set as d**3**, a depth of the laser irradiation mark is set as d**4**, and a volume occupancy rate V of the laser irradiation marks is set as (1/d**1**×d**3**×d**4**)×(1/d**2**)×100, the volume occupancy rate V of the laser irradiation marks is 0.00057% or more and 0.010% or less.

8 Claims, 3 Drawing Sheets

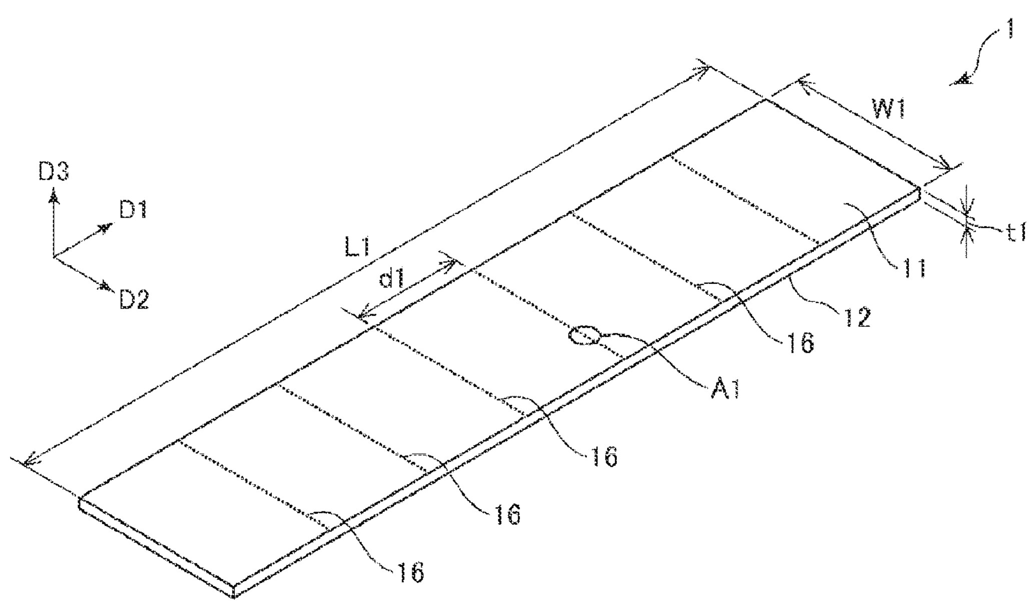

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FIG. 1

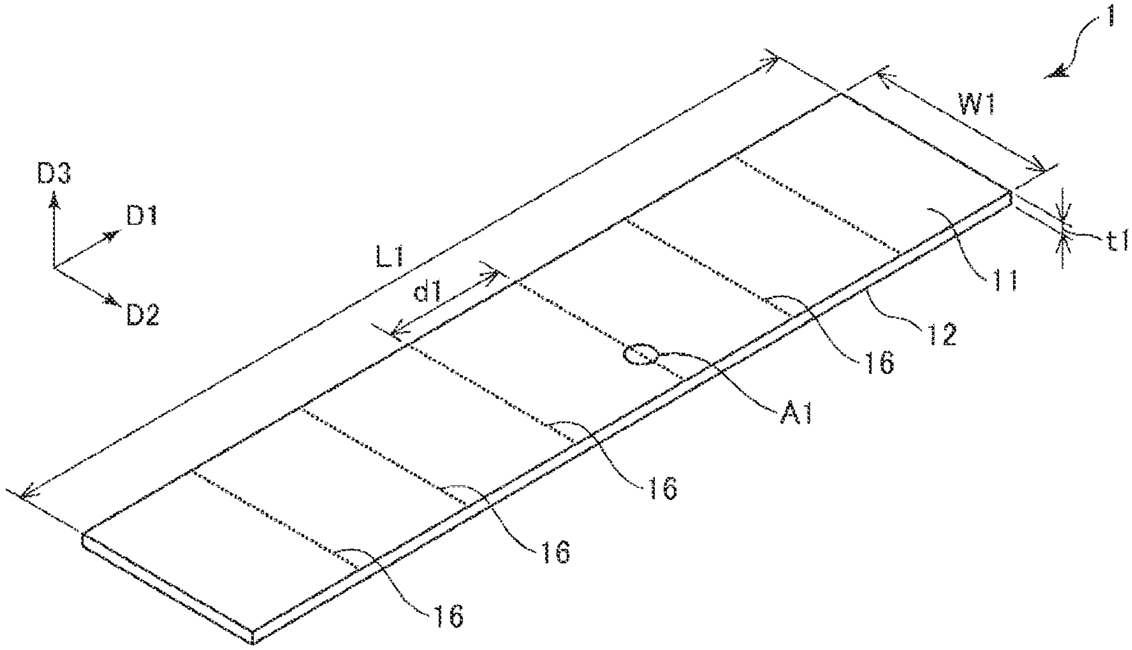


FIG. 2

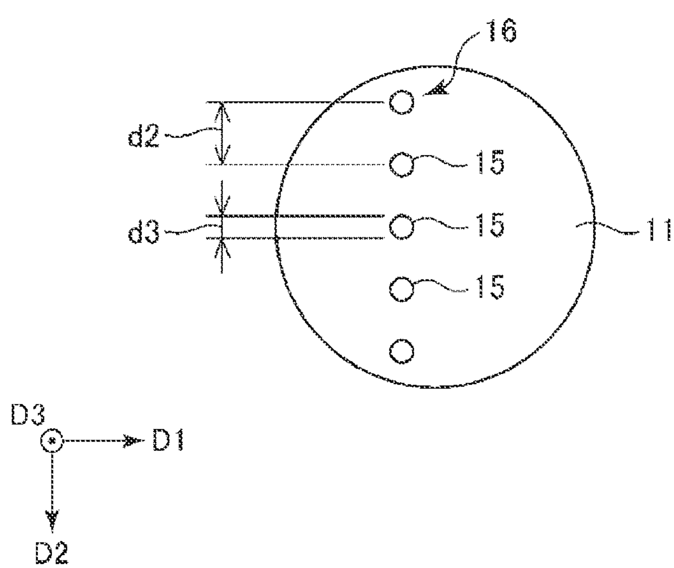


FIG. 3

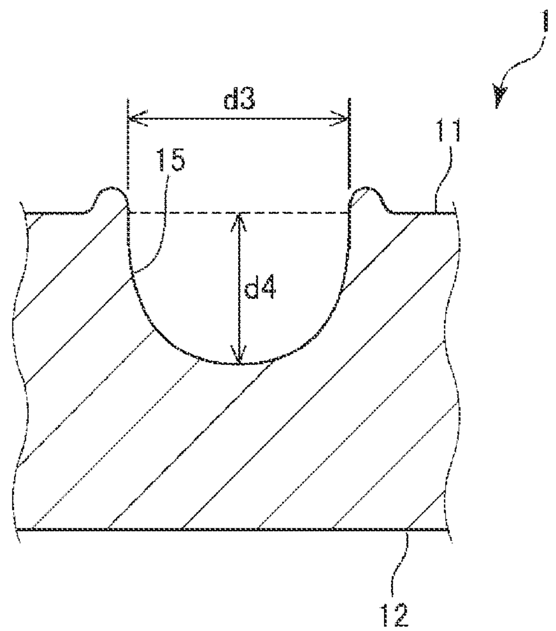

1
d3
11
15
d4
12

FIG. 4

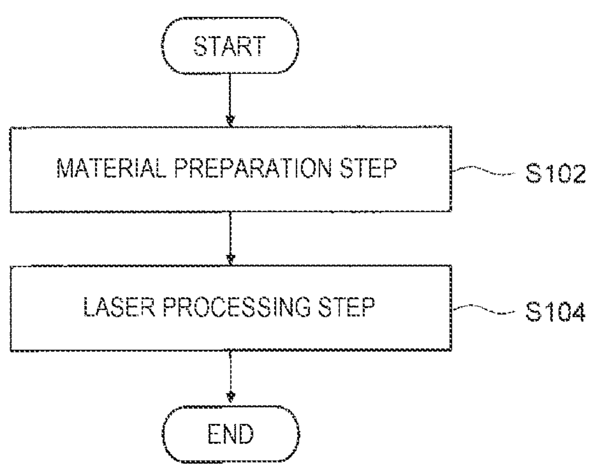
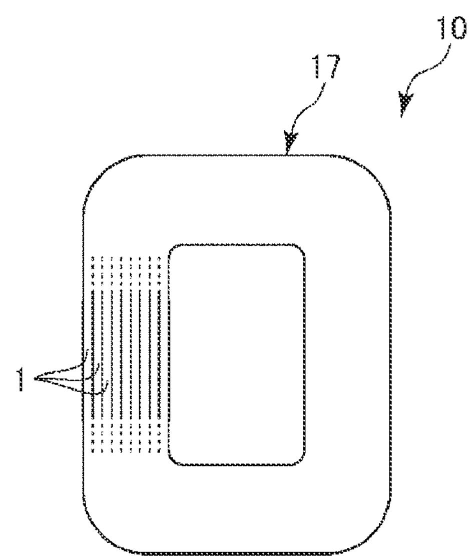

START
MATERIAL PREPARATION STEP
S102
LASER PROCESSING STEP
S104
END

FIG. 5



AMORPHOUS METAL RIBBON, METHOD FOR MANUFACTURING AMORPHOUS METAL RIBBON, AND MAGNETIC CORE

The present application is based on, and claims priority from JP Application Serial Number 2021-024949, filed Feb. 19, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to an amorphous metal ribbon, a method for manufacturing an amorphous metal ribbon, and a magnetic core.

2. Related Art

WO 2019/189813 discloses a Fe-based amorphous alloy ribbon having a plurality of laser irradiation mark rows each including a plurality of laser irradiation marks. In the Fe-based amorphous alloy ribbon, a distance (line distance d**1**) between the laser irradiation mark rows is 10 mm to 60 mm, and a distance (spot distance d**2**) between the laser irradiation marks is 0.10 mm to 0.50 mm. A number density D of the laser irradiation marks is represented by D=(1/d**1**)×(1/d**2**), and is 0.05 pieces/mm^2 to 0.50 pieces/mm^2.

In such a Fe-based amorphous alloy ribbon, an iron loss at an excitation magnetic flux density of 1.45 T is low, and an increase in excitation power is prevented. Accordingly, an iron core and a transformer having excellent performance can be realized.

In the Fe-based amorphous alloy ribbon described in WO 2019/189813, by controlling the line distance d**1**, the spot distance d**2**, and the number density D of the laser irradiation marks, magnetic domain refinement can be optimized, and the iron loss and the excitation power can be reduced. However, there is room for further improvement in magnetic domain refinement of the Fe-based amorphous alloy ribbon.

SUMMARY

An amorphous metal ribbon according to an application example of the present disclosure includes: a plurality of laser irradiation mark rows each including a plurality of laser irradiation marks arranged in a row, in which when a distance between the laser irradiation mark rows that are adjacent to each other is set as d**1**, a distance between the laser irradiation marks in the laser irradiation mark row is set as d**2**, a diameter of the laser irradiation mark is set as d**3**, a depth of the laser irradiation mark is set as d**4**, and a volume occupancy rate V of the laser irradiation marks is set as (1/d**1**×d**3**×d**4**)×(1/d**2**)×100, the volume occupancy rate V of the laser irradiation marks is 0.00057% or more and 0.010% or less.

A method for manufacturing an amorphous metal ribbon according to an application example of the present disclosure includes: preparing a material ribbon having a main surface made of an amorphous metal; applying laser processing to the main surface of the material ribbon to obtain an amorphous metal ribbon having a plurality of laser irradiation mark rows each including a plurality of laser irradiation marks arranged in a row, in which when a distance between the laser irradiation mark rows that are adjacent to each other is set as d**1**, a distance between the laser irradiation marks in the laser irradiation mark row is set as d**2**, a diameter of the laser irradiation mark is set as d**3**, a depth of the laser irradiation mark is set as d**4**, and a volume occupancy rate V of the laser irradiation marks is set as (1/d**1**×d**3**×d**4**)×(1/d**2**)×100, the volume occupancy rate V of the laser irradiation marks is 0.00057% or more and 0.010% or less.

A magnetic core according to an application example of the present disclosure includes: the amorphous metal ribbon according to the application example of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view schematically showing an amorphous metal ribbon according to an embodiment.

FIG. **2** is an enlarged view of a part A**1** in FIG. **1**.

FIG. **3** is a cross-sectional view of a laser irradiation mark shown in FIG. **2**.

FIG. **4** is a flowchart showing a method for manufacturing the amorphous metal ribbon according to the embodiment.

FIG. **5** is a schematic view showing a magnetic core according to the embodiment.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

Hereinafter, an amorphous metal ribbon, a method for manufacturing an amorphous metal ribbon, and a magnetic core according to the present disclosure will be described in detail based on preferred embodiments shown in the accompanying drawings.

1. Amorphous Metal Ribbon

An amorphous metal ribbon according to an embodiment is a ribbon made of an amorphous metal. The amorphous metal is a metal having no crystal structure. For example, a plurality of amorphous metal ribbons are stacked to form a laminate. Such a laminate is used as a magnetic core. The magnetic core is used in, for example, a transformer.

FIG. **1** is a perspective view schematically showing an amorphous metal ribbon according to the embodiment. FIG. **2** is an enlarged view of a part A**1** in FIG. **1**. In FIG. **1**, a length direction of an amorphous metal ribbon **1** is set as D**1**, a width direction thereof is set as D**2**, and a thickness direction thereof is set as D**3**. In FIG. **1**, the three directions are indicated by arrows. Each direction described below includes both a direction from a proximal end to a distal end of the arrow and a direction from the distal end to the proximal end of the arrow.

In FIG. **1**, a length of the amorphous metal ribbon **1** is set as L**1**, a width thereof is set as W**1**, and a thickness thereof is set as t**1**.

The ribbon has a first surface **11** and a second surface **12** having a front and back relation with each other, and has a shape that a thickness, which is a distance between the first surface **11** and the second surface **12**, is sufficiently shorter than a width and a length of the first surface **11** and the second surface **12**. The width of the first surface and the second surface, that is, the width W**1** of the amorphous metal ribbon **1**, is determined by a device or method for manufacturing the amorphous metal ribbon **1**. Therefore, the width is not particularly limited, and is preferably 5 mm or more, more preferably 10 mm or more and 200 mm or less, and still more preferably 20 mm or more and 100 mm or less.

The length of the first surface **11** and the second surface **12**, that is, the length L**1** of the amorphous metal ribbon **1**, is determined when the amorphous metal ribbon **1** is manufactured.

Therefore, the length is not particularly limited, and it is sufficient that the length is longer than the width W**1** of the amorphous metal ribbon **1**. As an example, the length L**1** of the amorphous metal ribbon is preferably 1.5 times or more, and more preferably 2.0 times or more the width W**1** of the amorphous metal ribbon **1**.

Examples of the amorphous metal include Fe-based amorphous alloys such as Fe-Si-B based, Fe-Si-B-C based, Fe-Si-B-Cr-C based, Fe-Si-Cr based, Fe-B based, Fe-B-C based, Fe-P-C based, Fe-Co-Si-B based, Fe-Si-B-Nb based, and Fe-Zr-B based amorphous alloys, Ni-based amorphous alloys such as Ni-Si-B based and Ni-P-B based amorphous alloys, and Co-based amorphous alloys such as a Co-Si-B based amorphous alloy. Among these, the Fe-based amorphous alloys are excellent in soft magnetism and have a high saturation magnetic flux density, and thus are useful as constituent materials of the amorphous metal ribbon **1** used for a magnetic core or the like.

The thickness t**1** of the amorphous metal ribbon **1**, that is, a distance between the first surface **11** and the second surface **12** in the thickness direction D**3**, is not particularly limited, and is preferably 10 μm or more and 50 μm or less, and more preferably 20 μm or more and 35 μm or less. The amorphous metal ribbon **1** having such a thickness t**1** achieves both sufficient mechanical strength and reduction in eddy current loss.

As shown in FIGS. **1** and **2**, the amorphous metal ribbon **1** according to the embodiment has laser irradiation mark rows **16** provided on the first surface **11**, each including a plurality of laser irradiation marks **15** arranged in a row.

FIG. **3** is a cross-sectional view of the laser irradiation mark **15** shown in FIG. **2**.

As shown in FIG. **2**, in the laser irradiation mark row **16**, the laser irradiation marks **15** that are substantially circular shapes in a plan view are arranged in a row along the width direction D**2** at a predetermined distance. That is, the laser irradiation mark row **16** is an aggregate of the laser irradiation marks **15** arranged in a row along the width direction D**2**. The laser irradiation mark **15** refers to a processing mark formed by irradiating the first surface **11** with a laser beam, and a processing mark at which the amorphous metal is melted by receiving the energy of the laser beam to form a concave shape as shown in FIG. **3**.

By providing such laser irradiation marks **15** to satisfy a predetermined arrangement condition, magnetic domain refinement is optimized in the amorphous metal ribbon **1**. Accordingly, in particular, it is possible to reduce an abnormal eddy current loss caused by the movement of a magnetic domain wall, which is a boundary between magnetic domains, and to reduce excitation power.

1.1. Line Distance

The distance between the laser irradiation mark rows **16** shown in FIG. **1** is set as a line distance d**1**. The line distance d**1** is preferably 10 mm or more and 60 mm or less, more preferably 10 mm or more and 50 mm or less, and still more preferably 10 mm or more and 30 mm or less. When the line distance d**1** is within the above range, an arrangement density of the laser irradiation mark rows **16** in the amorphous metal ribbon **1** can be optimized. As a result, even when a measurement is performed under a condition of a high magnetic flux density, an iron loss and the excitation power of the amorphous metal ribbon **1** can be reduced.

When the line distance d**1** is less than the lower limit value, the excitation power of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may increase depending on other arrangement conditions of the laser irradiation marks **15** and the laser irradiation mark rows **16**. When the line distance d**1** is more than the upper limit value, the iron loss of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may increase depending on other arrangement conditions of the laser irradiation marks **15** and the laser irradiation mark rows **16**.

The laser irradiation mark rows **16** that are adjacent to each other are preferably substantially parallel to each other, and may be non-parallel to each other. A portion in which the laser irradiation mark rows **16** are parallel to each other and a portion in which the laser irradiation mark rows **16** are non-parallel to each other may be mixed. The expression that the laser irradiation mark rows **16** are “parallel to each other” indicates that an angle between the laser irradiation mark rows **16** is 10° or less.

Each laser irradiation mark row **16** shown in FIG. **1** is substantially parallel to the width direction D**2**, and may be non-parallel to the width direction D**2**. A portion in which the laser irradiation mark **16** is parallel to the width direction D**2** and a portion in which the laser irradiation mark **16** is non-parallel to the width direction D**2** may be mixed. The expression that the laser irradiation mark row **16** is “parallel to” the width direction D**2** indicates that an angle between the laser irradiation mark row **16** and the width direction D**2** is 10° or less.

The line distance d**1** is a distance between centers of the laser irradiation marks **15**, which is measured in a middle portion of the width W**1** of the amorphous metal ribbon **1**. The middle portion refers to a region having a width half the width W**1** with the middle point of the width W**1** as the center. Therefore, when at least a part of the laser irradiation mark row **16** is provided in the middle portion, the laser irradiation mark row **16** may extend over the entire width W**1** of the amorphous metal ribbon **1** or may extend over only a part of the width W**1**. That is, the laser irradiation mark row **16** may be partially interrupted, or the length of the laser irradiation mark row **16** in the width direction D**2** may be shorter than the width W**1** of the amorphous metal ribbon **1**.

The distance between the laser irradiation mark rows **16** may be constant or may be partially different in the entire amorphous metal ribbon **1**. That is, when the distance between the laser irradiation mark rows **16** is measured at a plurality of positions in the middle portion of one width W**1** of the amorphous metal ribbon **1**, a plurality of measured values may be the same as or may be different from each other. In the latter case, an average value of five measured values is set as the line distance d**1** of the amorphous metal ribbon **1**.

The laser irradiation marks **15** may be provided in only one of the first surface **11** and the second surface **12**, or may be provided in both the first surface **11** and the second surface **12**. When the laser irradiation marks **15** are provided on both the first surface **11** and the second surface **12**, it is sufficient that the above range of the line distance d**1** is satisfied in a state where the laser irradiation marks **15** provided on the second surface **12** are projected onto the first surface **11** and the projected laser irradiation marks **15** and the laser irradiation marks **15** provided on the first surface **11** match each other.

1.2. Spot Distance

The distance between the laser irradiation marks **15** in the laser irradiation mark row **16** shown in FIG. **2** is set as a spot distance d**2**. The spot distance d**2** is preferably 0.10 mm or more and 0.50 mm or less, more preferably 0.15 mm or more and 0.40 mm or less, and still more preferably 0.20 mm or more and 0.40 mm or less. When the spot distance d**2** is within the above range, the arrangement density of the laser irradiation marks **15** in the laser irradiation mark row **16** can be optimized. As a result, even when the measurement is performed under a condition of a high magnetic flux density, the iron loss and the excitation power of the amorphous metal ribbon **1** can be reduced.

When the spot distance d**2** is less than the lower limit value, the excitation power of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may increase depending on other arrangement conditions of the laser irradiation marks **15** and the laser irradiation mark rows **16**. When the spot distance d**2** is more than the upper limit value, the iron loss of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may increase depending on other arrangement conditions of the laser irradiation marks **15** and the laser irradiation mark rows **16**.

The spot distance d**2** is a distance between centers of the laser irradiation marks **15**, which is measured in a middle portion of the width W**1** of the amorphous metal ribbon **1**. The center of each laser irradiation mark **15** is the center of a perfect circle inscribed in the laser irradiation mark **15**.

The distance between the laser irradiation marks **15** may be constant or may be partially different in the entire amorphous metal ribbon **1**. That is, when the distance between the laser irradiation marks **15** is measured at a plurality of positions in the middle portion of one width W**1** of the amorphous metal ribbon **1**, a plurality of measured values may be the same as or may be different from each other. In the latter case, an average value of five measured values is set as the spot distance d**2** of the amorphous metal ribbon **1**.

When the laser irradiation marks **15** are provided on both the first surface **11** and the second surface **12**, it is sufficient that the above range of the spot distance d**2** is satisfied in a state where the laser irradiation marks **15** provided on the second surface **12** are projected onto the first surface **11** and the projected laser irradiation marks **15** and the laser irradiation marks **15** provided on the first surface **11** match each other.

1.3. Spot Diameter

An equivalent circle diameter of the laser irradiation mark **15** shown in FIG. **2** is set as a spot diameter d**3**. The spot diameter d**3** is preferably 0.010 mm or more and 0.30 mm or less, more preferably 0.020 mm or more and 0.25 mm or less, and still more preferably 0.030 mm or more and 0.20 mm or less. When the spot diameter d**3** is within the above range, the magnetic domain refinement based on the laser irradiation marks **15** can be efficiently performed. In addition, it is possible to prevent a decrease in mechanical strength of the amorphous metal ribbon **1** caused by the formation of the laser irradiation marks **15**.

When the spot diameter d**3** is less than the lower limit value, depending on other arrangement conditions of the laser irradiation marks **15** and the laser irradiation mark rows **16**, the magnetic domain refinement may be insufficient and the iron loss and the excitation power of the amorphous metal ribbon **1**, which are measured under the condition of a high magnetic flux density, may increase. When the spot diameter d**3** is more than the upper limit value, the mechanical strength of the amorphous metal ribbon **1** may decrease.

The spot diameter d**3** is an average value of the circle equivalent diameters of 10 or more laser irradiation marks **15**, which are measured in a middle portion of the width W**1** of the amorphous metal ribbon **1**. The equivalent circle diameter is a diameter of a perfect circle having an area same as that of the laser irradiation mark **15** when the first surface **11** is viewed in a plan view.

The equivalent circle diameters of the laser irradiation marks **15** may be the same as or different from each other. However, considering the influence of the laser irradiation marks **15** on the mechanical strength of the amorphous metal ribbon **1**, it is preferable that the equivalent circle diameters of the laser irradiation marks **15** have a slight variation. Accordingly, when the amorphous metal ribbon **1** is curved, it is possible to prevent local concentration of stress. As a result, even when the laser irradiation marks **15** are arranged at a high density, the mechanical strength of the amorphous metal ribbon **1** is less likely to decrease.

Specifically, first, in the middle portion of the width W**1** of the amorphous metal ribbon **1**, a standard deviation and an average value of the equivalent circle diameters of 100 laser irradiation marks **15** are calculated. Next, a variation coefficient is calculated by dividing the standard deviation by the average value. The variation coefficient is an index indicating the degree of variation in equivalent circle diameters of the laser irradiation marks **15**, and the larger the variation coefficient, the larger the degree of variation. In the amorphous metal ribbon **1**, the variation coefficient is preferably 3% or more, more preferably 3% or more and 50% or less, and still more preferably 8% or more and 30% or less. By setting the degree of variation in equivalent circle diameters of the laser irradiation marks **15** within the above range, stress concentration relaxation along with an appropriate variation can be achieved, the decrease in mechanical strength of the amorphous metal ribbon **1** can be prevented, and the laser irradiation marks **15** can be arranged at a high density. As a result, it is possible to increase the degree of freedom of arrangement of the laser irradiation marks **15**, and it is possible to sufficiently reduce the iron loss and the excitation power of the amorphous metal ribbon **1**, which are measured under a condition of a high magnetic flux density.

In a case where the variation coefficient is less than the lower limit value, when the amorphous metal ribbon **1** is curved, the stress may be locally concentrated, and the mechanical strength of the amorphous metal ribbon **1** may be reduced. The variation coefficient may be more than the upper limit value, but depending on other arrangement conditions of the laser irradiation marks **15** and the laser irradiation mark rows **16**, the excitation power of the amorphous metal ribbon **1** may increase, or the iron loss may not be sufficiently prevented, which are measured under the condition of a high magnetic flux density.

1.4. Spot Depth

The depth of the laser irradiation mark **15** shown in FIG. **3** is set as a spot depth d**4**. The spot depth d**4** is preferably 0.0020 mm or more and 0.15 mm or less, more preferably 0.0030 mm or more and 0.10 mm or less, and still more preferably 0.0040 mm or more and 0.050 mm or less. When the spot depth d**4** is within the above range, the magnetic domain refinement based on the laser irradiation marks **15** can be necessarily and sufficiently performed. In addition, it is possible to prevent a decrease in mechanical strength of the amorphous metal ribbon **1** caused by the formation of the laser irradiation marks **15**.

When the spot depth d4 is less than the lower limit value, depending on other arrangement conditions of the laser irradiation marks **15** and the laser irradiation mark rows **16**, the magnetic domain refinement may be insufficient and the iron loss and the excitation power of the amorphous metal ribbon **1**, which are measured under the condition of a high magnetic flux density, may increase. When the spot depth d**4** is more than the upper limit value, the mechanical strength of the amorphous metal ribbon **1** may decrease.

The spot depth d**4** is an average value of the depths of 10 or more laser irradiation marks **15**, which are measured in a middle portion of the width W**1** of the amorphous metal ribbon **1**.

The depths of the laser irradiation marks **15** may be the same as or different from each other.

1.5. Number Density of Laser Irradiation Marks

A number density D of the laser irradiation marks **15** can be represented by using the line distance d**1** [mm] and the spot distance d**2** [mm] in the amorphous metal ribbon **1**. Specifically, the number density D of the laser irradiation marks **15** is represented by (1/d**1**)×(1/d**2**). The number density D is an index indicating the arrangement density based on the number of the laser irradiation marks **15**. The number density D in the amorphous metal ribbon **1** is preferably 0.05 pieces/mm^2 or more and 0.50 pieces/mm^2 or less, more preferably 0.10 pieces/mm^2 or more and 0.40 pieces/mm^2 or less, and still more preferably 0.15 pieces/mm^2 or more and 0.35 pieces/mm^2 or less. When the number density D is within the above range, the magnetic domain refinement based on the laser irradiation marks **15** can be optimized, and even when the measurement is performed under a condition of a high magnetic flux density, the iron loss and the excitation power of the amorphous metal ribbon **1** can be reduced.

When the number density D is less than the lower limit value, the iron loss of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may not be sufficiently reduced. When the number density D is more than the upper limit value, an increase in excitation power of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may not be sufficiently prevented. When the amorphous metal ribbon **1** is curved, damages such as breakage may be likely to occur in the amorphous metal ribbon **1**.

The number density D is calculated based on a region in which the laser irradiation mark rows **16** are arranged and which has a length in the length direction D**1** of 1 m or more, in the middle portion of the width W**1** of the amorphous metal ribbon **1**. When the length L**1** of the amorphous metal ribbon **1** is less than 1 m, the number density D is calculated based on the total length.

When the laser irradiation marks **15** are provided on both the first surface **11** and the second surface **12**, it is sufficient that the above range of the number density D is satisfied in a state where the laser irradiation marks **15** provided on the second surface **12** are projected onto the first surface **11** and the projected laser irradiation marks **15** and the laser irradiation marks **15** provided on the first surface **11** match each other.

1.6. Volume Occupancy Rate of Laser Irradiation Marks

A volume occupancy rate V of the laser irradiation marks **15** can be represented by using the line distance d**1**, the spot distance d**2**, the spot diameter d**3**, and the spot depth d**4** in the amorphous metal ribbon **1**. Specifically, the volume occupancy rate V (%) of the laser irradiation marks **15** is represented by (1/d**1**×d**3**×d**4**)×(1/d**2**)×100. The volume occupancy rate V is an index indicating the arrangement density based on a volume of the laser irradiation marks **15**. The volume occupancy rate V in the amorphous metal ribbon **1** is 0.00057% or more and 0.010% or less, preferably 0.0010% or more and 0.0080% or less, and more preferably 0.0030% or more and 0.0057% or less. When the volume occupancy rate V is within the above range, the magnetic domain refinement based on the laser irradiation marks **15** can be optimized, and even when the measurement is performed under a condition of a high magnetic flux density, the iron loss and the excitation power of the amorphous metal ribbon **1** can be reduced. When the volume occupancy rate V is within the above range, the influence of the laser irradiation marks **15** on the mechanical strength of the amorphous metal ribbon **1** can be prevented to be small. Therefore, even when the amorphous metal ribbon **1** is curved, breakage or the like of the amorphous metal ribbon **1** is less likely to occur, and a good curved state can be maintained.

When the volume occupancy rate V is less than the lower limit value, the iron loss of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may not be sufficiently reduced. When the volume occupancy rate V is more than the upper limit value, an increase in excitation power of the amorphous metal ribbon **1**, which is measured under the condition of a high magnetic flux density, may not be sufficiently prevented. When the amorphous metal ribbon **1** is curved, damages such as breakage may be likely to occur in the amorphous metal ribbon **1**.

The volume occupancy rate V is calculated based on a region in which the laser irradiation mark rows **16** are arranged and which has a length in the length direction D**1** of 1 m or more, in the middle portion of the width W**1** of the amorphous metal ribbon **1**. When the length L**1** of the amorphous metal ribbon **1** is less than 1 m, the volume occupancy rate V is calculated based on the total length.

When the laser irradiation marks **15** are provided on both the first surface **11** and the second surface **12**, it is sufficient that the above range of the volume occupancy rate V is satisfied in a state where the laser irradiation marks **15** provided on the second surface **12** are projected onto the first surface **11** and the projected laser irradiation marks **15** and the laser irradiation marks **15** provided on the first surface **11** match each other.

As described above, the amorphous metal ribbon **1** according to the present embodiment includes a plurality of the laser irradiation mark rows **16** each including a plurality of the laser irradiation marks **15** arranged in a row. In the amorphous metal ribbon **1**, a distance between the laser irradiation mark rows **16** that are adjacent to each other is set as the line distance d**1**, a distance between the laser irradiation marks **15** in the laser irradiation mark row **16** is set as the spot distance d**2**, a diameter of the laser irradiation mark **15** is set as the spot diameter d**3**, and a depth of the laser irradiation mark **15** is set as the spot depth d**4**.

In the amorphous metal ribbon **1**, when the volume occupancy rate V of the laser irradiation marks **15** is set as (1/d**1**×d**3**×d**4**)×(1/d**2**)×100, the volume occupancy rate V of the laser irradiation marks **15** is 0.00057% or more and 0.010% or less.

According to such a configuration, the arrangement density of the laser irradiation marks **15** based on the volume can be optimized, and the magnetic domain refinement based on the laser irradiation marks **15** can be optimized. As a result, even when the measurement is performed at a relatively high magnetic flux density of, for example, 1.45 T, the iron loss and the excitation power of the amorphous metal ribbon **1** can be reduced. Therefore, the amorphous metal ribbon **1** according to the present embodiment can contribute to, for example, when the amorphous metal ribbon **1** is used in a transformer, miniaturization of the transformer and high efficiency of the transformer.

Even when such an amorphous metal ribbon **1** is curved, it is possible to prevent the damages such as breakage from occurring. Therefore, the amorphous metal ribbon **1** can be curved with a small bending radius, and miniaturization of the magnetic core can be achieved. From this viewpoint, the amorphous metal ribbon **1** also contributes to the miniaturization of the transformer or the like.

As described above, in the amorphous metal ribbon **1** according to the present embodiment, the line distance d**1** between the laser irradiation mark rows **16** is preferably 10 mm or more and 60 mm or less. The spot distance d**2** between the laser irradiation marks **15** is preferably 0.10 mm or more and 0.50 mm or less. The spot diameter d**3** of the laser irradiation mark **15** is preferably 0.010 mm or more and 0.30 mm or less.

According to such a configuration, the magnetic domain refinement based on the laser irradiation marks **15** can be optimized. As a result, the iron loss and the excitation power of the amorphous metal ribbon **1** can be sufficiently reduced.

1.7. Alloy Composition

As described above, the amorphous metal constituting the amorphous metal ribbon **1** is preferably a Fe-based amorphous alloy.

As the Fe-based amorphous alloy, a Fe-Si-B-based alloy or a Fe-Si-B-C-based alloy among the alloys described above is particularly preferably used. Among these, the Fe-Si-B-based alloy is made of Fe, Si, B, and impurities. The Fe-Si-B-based alloy has a chemical composition in which, when the total content of Fe, Si, and B is 100 atomic %, a Fe content is 78 atomic % or more, a B content is 11 atomic % or more, and the total content of Si and B is 17 atomic % or more and 22 atomic % or less.

Fe is a metal element having a large magnetic moment, and determines the magnetic flux density of the amorphous metal ribbon **1**. The Fe content is preferably 80 atomic % or more and 82 atomic % or less.

Si and B determine an amorphous-forming ability of the Fe-based amorphous alloy. A Si content is preferably 2.0 atomic % or more and 6.0 atomic % or less, and more preferably 3.5 atomic % or more and 6.0 atomic % or less. The B content is preferably 12 atomic % or more and 16 atomic % or less, and more preferably 13 atomic % or more and 16 atomic % or less. As described above, the total content of Si and B is preferably 17 atomic % or more and 22 atomic % or less.

In the Fe-based amorphous alloy having such a chemical composition, in particular, by setting the Fe content within the above range, it is possible to improve the magnetic flux density while improving the amorphous-forming ability. Therefore, it is possible to obtain the amorphous metal ribbon **1** that exhibits excellent soft magnetism derived from amorphous substances and has a high saturation magnetic flux density. In particular, by setting the total content of Si and B within the above range, it is possible to obtain the amorphous metal ribbon **1** in which the iron loss and the excitation power are sufficiently reduced.

1.8. Iron Loss and Excitation Power

In the amorphous metal ribbon **1** according to the present embodiment, as described above, the iron loss and the excitation power that are measured under the condition of a high magnetic flux density can be particularly reduced.

Specifically, the iron loss of the amorphous metal ribbon **1** under a condition of a frequency of 60 Hz and a magnetic flux density of 1.45 T is preferably 0.150 W/kg or less, more preferably 0.140 W/kg or less, and still more preferably 0.130 W/kg or less.

The excitation power of the amorphous metal ribbon **1** under the condition of a frequency of 60 Hz and a magnetic flux density of 1.45 T is preferably 0.180 VA/kg or less, more preferably 0.170 VA/kg or less, and still more preferably 0.165 VA/kg or less.

When the amorphous metal ribbon **1** satisfying such iron loss and excitation power is used for, for example, a transformer, the amorphous metal ribbon **1** greatly contributes to high efficiency of the transformer.

1.9. Magnetic Flux Density

In the amorphous metal ribbon **1** according to the present embodiment, it is possible to reduce the excitation power measured under the condition of a high magnetic flux density, and therefore, it is possible to prevent a decrease in magnetic flux density caused by an increase in excitation power. Therefore, it is possible to realize a high magnetic flux density in the amorphous metal ribbon **1**. Specifically, in the amorphous metal ribbon **1**, the magnetic flux density under a condition of a frequency of 60 Hz and a magnetic field of 7.9557 A/m is preferably 1.52 T or more, and more preferably 1.62 T or more. When such an amorphous metal ribbon **1** is used for, for example, a transformer, the amorphous metal ribbon **1** contributes to the miniaturization of the transformer.

2. Method for Manufacturing Amorphous Metal Ribbon

Next, a method for manufacturing the amorphous metal ribbon according to the embodiment will be described.

FIG. **4** is a flowchart showing the method for manufacturing the amorphous metal ribbon according to the embodiment.

The method for manufacturing the amorphous metal ribbon shown in FIG. **4** includes a material preparation step S**102** and a laser processing step S**104**. In the material preparation step S**102**, a material ribbon made of an amorphous metal is prepared. In the laser processing step S**104**, laser processing is performed on one of two main surfaces of the material ribbon that are in a front and back relation, and an amorphous metal ribbon having a plurality of laser irradiation mark rows each including a plurality of laser irradiation marks arranged in a row is obtained.

In the amorphous metal ribbon, a distance between the laser irradiation mark rows that are adjacent to each other is set as the line distance d**1**, a distance between the laser irradiation marks in the laser irradiation mark row is set as the spot distance d**2**, a diameter of the laser irradiation mark is set as the spot diameter d**3**, and a depth of the laser irradiation mark is set as the spot depth d**4**.

In the amorphous metal ribbon, when the volume occupancy rate V of the laser irradiation marks is set as $(1/d1\times d3\times d4)\times(1/d2)\times100$, the volume occupancy rate V of the laser irradiation marks is 0.00057% or more and 0.010% or less.

According to such a configuration, the arrangement density of the laser irradiation marks of the manufactured amorphous metal ribbon based on the volume can be optimized, and the magnetic domain refinement based on the laser irradiation marks can be optimized. As a result, it is possible to efficiently manufacture an amorphous metal ribbon in which the iron loss and the excitation power under the condition of a high magnetic flux density are reduced. Even when the amorphous metal ribbon is curved, it is possible to efficiently manufacture an amorphous metal ribbon in which damages such as breakage are less likely to occur.

2.1. Material Preparation Step

The material ribbon is manufactured by, for example, a method for manufacturing a rapidly solidified ribbon, such as a single roll method. The material preparation step **S102** may be a step of manufacturing the material ribbon by such a manufacturing method, may include a step of cutting the material ribbon manufactured by the manufacturing method described above into a necessary length, or may be a step of only preparing the material ribbon.

2.2. Laser Processing Step

In the laser processing step **S104**, laser processing is performed on at least one of two main surfaces of the material ribbon, and laser irradiation marks are formed. The arrangement and the like of the laser irradiation marks are the same as the arrangement and the like of the laser irradiation marks **15** in the amorphous metal ribbon **1** described above.

Conditions of the laser processing vary depending on the alloy composition and the like of the material ribbon. As an example, an output of laser in the laser processing is preferably 0.4 mJ or more and 2.5 mJ or less, and more preferably 1.0 mJ or more and 2.0 mJ or less.

A diameter of a laser beam in the laser processing determines the spot diameter d**3** described above. As an example, the diameter of the laser beam is preferably 0.010 mm or more and 0.30 mm or less, and more preferably 0.020 mm or more and 0.25 mm or less.

An energy density of the laser in the laser processing determines the spot diameter d**3** and the depth of the laser irradiation mark described above. As an example, the energy density of the laser is preferably 0.01 J/mm^2 or more and 1.50 J/mm^2 or less, and more preferably 0.03 J/mm^2 or more and 1.00 J/mm^2 or less.

A wavelength of the laser in the laser processing is, for example, preferably 250 nm or more and 1100 mm or less, and more preferably 900 nm or more and 1100 nm or less.

A laser light source in the laser processing includes, for example, a YAG laser, a CO_2 gas laser, a semiconductor laser, and a fiber laser. Among these, from the viewpoint of being capable of emitting high-frequency pulsed laser light with high output, the fiber laser is preferably used. A pulse width of the pulsed laser light is preferably 50 nanoseconds or more, and more preferably 100 nanoseconds or more. The pulse width is a time during which laser irradiation is performed, and when the pulse width is small, the irradiation time is shorter. By setting the pulse width within the above range, a laser irradiation mark having an appropriate size and depth can be efficiently formed.

3. Magnetic Core

Next, a magnetic core according to the embodiment will be described.

FIG. **5** is a schematic view showing the magnetic core according to the embodiment.

A magnetic core **10** shown in FIG. **5** is made of a laminate **17** obtained by laminating a plurality of amorphous metal ribbons **1**. Specifically, an annular magnetic core **10** shown in FIG. **5** is formed by curving the laminate **17** and overlapping and wrapping both ends of the laminate **17**. A well-known overlapping and wrapping method is used as the overlapping and wrapping method.

A shape of the magnetic core **10** is not limited to the shape shown in FIG. **5**, and may be any shape.

The amorphous metal ribbons **1** provided in the laminate **17** are preferably insulated from each other. For the insulation, for example, a resin coating can be used.

As described above, the magnetic core **10** includes the amorphous metal ribbons **1** described above. Accordingly, even when the measurement is performed under a condition of a high magnetic flux density, a magnetic core **10** having low iron loss and excitation power can be obtained.

Such a magnetic core **10** is suitably used for, for example, a transformer, a saturated reactor, and a magnetic switch.

The amorphous metal ribbon, the method for manufacturing the amorphous metal ribbon, and the magnetic core according to the present disclosure have been described above based on preferred embodiments, but the present disclosure is not limited thereto.

EXAMPLES

Next, specific Examples of the present disclosure will be described.

4. Manufacturing of Amorphous Metal Ribbon

4.1. Example 1

First, a material ribbon made of an amorphous metal having an alloy composition of $Fe_{82}Si_4B_{14}$ and having a thickness of 25 μm and a width of 210 mm was manufactured by a single roll method. $Fe_{82}Si_4B_{14}$ means an alloy composition in which when the total content of Fe, Si, and B is 100 atomic %, the Fe content is 82 atomic %, the Si content is 4 atomic %, and the B content is 14 atomic %.

Next, a sample piece having a size of a length of 120 mm and a width of 25 mm was cut out from the manufactured material ribbon.

Next, the laser processing is performed on at least one main surface of the cut sample piece, and laser irradiation marks were formed. In this case, the spot diameters of the laser irradiation marks were adjusted by changing the energy density of the laser beam. As shown in FIG. **1**, the laser irradiation mark rows including the laser irradiation marks were formed over the entire width direction of the material ribbon. Table 1 shows arrangement conditions of the laser irradiation marks. Accordingly, an amorphous metal ribbon was obtained.

4.2. Examples 2 to 34

Amorphous metal ribbons were obtained in the same manner as in Example 1 except that the arrangement conditions of the laser irradiation marks were changed as shown in Table 1 or 2.

4.3. Comparative Examples 1 to 12

Amorphous metal ribbons were obtained in the same manner as in Example 1 except that the arrangement conditions of the laser irradiation marks were changed as shown in Table 1 or 2.

5. Evaluation of Amorphous Metal Ribbon

5.1. Iron Loss

For the amorphous metal ribbons obtained in Examples and Comparative Examples, the iron loss under the condition of a frequency of 60 Hz and a magnetic flux density of 1.45 T was measured. Next, measurement results were evaluated in light of the following evaluation criteria.

A: the iron loss is 0.120 W/kg or less

B: the iron loss is more than 0.120 W/kg and 0.130 W/kg or less

C: the iron loss is more than 0.130 W/kg and 0.140 W/kg or less

D: the iron loss is more than 0.140 W/kg and 0.150 W/kg or less

E: the iron loss is more than 0.150 W/kg and 0.160 W/kg or less

F: the iron loss is more than 0.160 W/kg

The evaluation results are shown in Tables 1 and 2.

5.2. Excitation Power

For the amorphous metal ribbons obtained in Examples and Comparative Examples, the excitation power under the condition of the frequency of 60 Hz and the magnetic flux density of 1.45 T was measured. Next, measurement results were evaluated in light of the following evaluation criteria.

A: the excitation power is 0.135 VA/kg or less

B: the excitation power is more than 0.135 VA/kg and 0.150 VA/kg or less

C: the excitation power is more than 0.150 VA/kg and 0.165 VA/kg or less

D: the excitation power is more than 0.165 VA/kg and 0.180 VA/kg or less

E: the excitation power is more than 0.180 VA/kg and 0.195 VA/kg or less

F: the excitation power is more than 0.195 VA/kg

The evaluation results are shown in Tables 1 and 2.

TABLE 1

	Arrangement condition of laser irradiation mark							Evaluation result		
	Line distance d1 mm	Spot distance d2 mm	Spot diameter d3 mm	Spot depth d4 mm	Number density D piece/mm^2	Volume occupancy rate V %	Variation coefficient of diameter of irradiation mark %	Iron loss —	Excitation power —	Bending resistance —
Comparative Example 1	10	0.10	0.030	0.006	1.000	0.0180	20	A	F	F
Example 1	20	0.10	0.030	0.006	0.500	0.0090	11	A	D	D
Example 2	30	0.10	0.030	0.006	0.333	0.0060	10	A	C	B
Example 3	40	0.10	0.030	0.006	0.250	0.0045	13	A	C	B
Example 4	50	0.10	0.030	0.006	0.200	0.0036	12	A	C	B
Example 5	60	0.10	0.030	0.006	0.167	0.0030	15	A	D	B
Example 6	10	0.20	0.030	0.006	0.500	0.0090	22	A	D	D
Example 7	20	0.20	0.030	0.006	0.250	0.0045	13	A	A	A
Example 8	30	0.20	0.030	0.006	0.167	0.0030	16	A	A	A
Example 9	40	0.20	0.030	0.006	0.125	0.0023	25	B	A	A
Example 10	50	0.20	0.030	0.006	0.100	0.0018	18	B	A	A
Example 11	60	0.20	0.030	0.006	0.083	0.0015	21	C	B	A
Example 12	10	0.30	0.030	0.006	0.333	0.0060	20	A	A	A
Example 13	20	0.30	0.030	0.006	0.167	0.0030	17	A	A	A
Example 14	30	0.30	0.030	0.006	0.111	0.0020	16	B	A	A
Example 15	40	0.30	0.030	0.006	0.083	0.0015	14	C	B	A
Example 16	50	0.30	0.030	0.006	0.067	0.0012	12	C	B	A
Example 17	60	0.30	0.030	0.006	0.056	0.0010	25	C	C	A
Example 18	10	0.40	0.025	0.006	0.250	0.0039	34	A	A	A
Example 19	20	0.40	0.025	0.006	0.125	0.0020	15	B	A	A
Example 20	30	0.40	0.025	0.006	0.083	0.0013	12	C	A	A
Example 21	40	0.40	0.025	0.006	0.063	0.0010	16	C	C	A
Example 22	50	0.40	0.025	0.006	0.050	0.0008	23	D	C	A
Example 23	60	0.40	0.025	0.006	0.042	0.0007	15	D	D	A
Example 24	10	0.50	0.025	0.006	0.200	0.0031	11	B	A	A
Example 25	20	0.50	0.025	0.006	0.100	0.0016	10	B	A	A
Example 26	30	0.50	0.025	0.006	0.067	0.0010	9	C	C	B
Example 27	40	0.50	0.025	0.006	0.050	0.0008	17	D	D	A
Comparative Example 2	50	0.50	0.025	0.005	0.040	0.0005	16	E	D	A
Comparative Example 3	60	0.50	0.025	0.006	0.033	0.0005	13	F	E	A

TABLE 2

	Arrangement condition of laser irradiation mark							Evaluation result		
	Line distance d1 mm	Spot distance d2 mm	Spot diameter d3 mm	Spot depth d4 mm	Number density D piece/mm^2	Volume occupancy rate V %	Variation coefficient of diameter of irradiation mark %	Iron loss —	Excitation power —	Bending resistance —
Comparative Example 4	50	0.40	0.015	0.003	0.050	0.0002	17	F	D	A

TABLE 2-continued

	Arrangement condition of laser irradiation mark							Evaluation result		
	Line distance d1 mm	Spot distance d2 mm	Spot diameter d3 mm	Spot depth d4 mm	Number density D piece/mm^2	Volume occupancy rate V %	Variation coefficient of diameter of irradiation mark %	Iron loss —	Excitation power —	Bending resistance —
Example 28	50	0.40	0.050	0.010	0.050	0.0025	16	B	B	B
Example 29	50	0.40	0.070	0.014	0.050	0.0049	20	A	B	A
Example 30	50	0.40	0.100	0.020	0.050	0.0100	23	A	D	D
Comparative Example 5	50	0.40	0.225	0.045	0.050	0.0506	19	A	E	F
Comparative Example 6	50	0.40	0.250	0.050	0.050	0.0625	24	A	F	F
Comparative Example 7	60	0.30	0.275	0.055	0.056	0.0840	13	F	D	A
Example 31	60	0.30	0.050	0.010	0.056	0.0028	16	A	B	A
Example 32	60	0.30	0.070	0.014	0.056	0.0054	24	A	B	A
Comparative Example 8	60	0.30	0.125	0.025	0.056	0.0174	16	A	E	E
Comparative Example 9	60	0.30	0.150	0.030	0.056	0.0250	22	A	E	F
Comparative Example 10	60	0.30	0.175	0.035	0.056	0.0340	12	A	F	F
Comparative Example 11	70	0.25	0.200	0.040	0.057	0.0457	20	B	F	F
Comparative Example 12	70	0.30	0.250	0.050	0.048	0.0595	18	B	F	F
Example 33	20	0.30	0.030	0.006	0.167	0.0030	2	A	A	D
Example 34	20	0.30	0.030	0.006	0.167	0.0030	62	A	D	A

As is clear from Tables 1 and 2, the amorphous metal ribbons obtained in Examples are lower in both iron loss and excitation power when the iron loss and the excitation power are measured at a relatively high magnetic flux density of 1.45 T as compared with the amorphous metal ribbons obtained in Comparative Examples.

5.3. Magnetic Flux Density

For the amorphous metal ribbons obtained in Examples and Comparative Examples, the magnetic flux density under the condition of a frequency of 60 Hz and a magnetic field of 7.9557 A/m was measured.

As a result, all of the amorphous metal ribbons obtained in Examples have a magnetic flux density of 1.52 T or more, and show a high magnetic flux density as compared with the amorphous metal ribbons obtained in Comparative Examples.

5.4. Bending Resistance

For the amorphous metal ribbons obtained in Examples and Comparative Examples, bending resistance was evaluated using a cylindrical mandrel testing machine specified in JIS K5600-5-1. Specifically, a test was performed in which the amorphous metal ribbon was bent along a core rod having a diameter of 0.73 mm and a bending angle was gradually reduced from 180° toward 0°. Next, the bending angle at which the amorphous metal ribbon was broken was recorded, and the bending angle was evaluated in light of the following evaluation criteria.

A: the bending angle at the time of breakage is 30° or less

B: the bending angle at the time of breakage is more than 30° and 60° or less

C: the bending angle at the time of breakage is more than 60° and 90° or less

D: the bending angle at the time of breakage is more than 90° and 120° or less

E: the bending angle at the time of breakage is more than 120° and 150° or less

F: the bending angle at the time of breakage is more than 150° and 180° or less

The evaluation results are shown in Table 1.

As shown in Table 1, it is confirmed that the amorphous metal ribbons obtained in Examples have a bending angle at which the amorphous metal ribbons are broken smaller than those of the amorphous metal ribbons obtained in Comparative Examples. Therefore, it is confirmed that the amorphous metal ribbons obtained in Examples have excellent bending resistance. Such amorphous metal ribbons contribute to, for example, the miniaturization of a magnetic core.

What is claimed is:

1. An amorphous metal ribbon, comprising:

a plurality of laser irradiation mark rows each including a plurality of laser irradiation marks arranged in a row, wherein

a distance between the plurality of laser irradiation mark rows that are adjacent to each other is set as d**1**,

a distance between the plurality of laser irradiation marks in each laser irradiation mark row of the plurality of laser irradiation mark rows is set as d**2**,

a diameter of a laser irradiation mark of the plurality of laser irradiation marks is set as d**3**,

a depth of the laser irradiation mark which is set as d**4** is 0.0055 mm or more and 0.15 mm or less, and

a volume occupancy rate V of the plurality of laser irradiation marks is set as $(1/d1 \times d3 \times d4) \times (1/d2) \times 100$, the volume occupancy rate V of the plurality of laser irradiation marks is 0.00057% or more and 0.010% or less.

2. The amorphous metal ribbon according to claim **1**, wherein

a number density D of the plurality of laser irradiation marks is 0.05 pieces/mm^2 or more and 0.50 pieces/mm^2 or less.

3. The amorphous metal ribbon according to claim **1**, wherein

the distance d**1** between the plurality of laser irradiation mark rows is 10 mm or more and 60 mm or less,

the distance d**2** between the plurality of laser irradiation marks is 0.10 mm or more and 0.50 mm or less, and

the diameter d**3** of the laser irradiation mark is 0.010 mm or more and 0.30 mm or less.

4. The amorphous metal ribbon according to claim **1**, wherein

a thickness is 10 μm or more and 50 μm or less.

5. The amorphous metal ribbon according to claim **1**, wherein

an iron loss under a condition of a frequency of 60 Hz and a magnetic flux density of 1.45 T is 0.150 W/kg or less, and

an excitation power under the condition of the frequency of 60 Hz and the magnetic flux density of 1.45 T is 0.180 VA/kg or less.

6. The amorphous metal ribbon according to claim **1**, wherein

a variation coefficient obtained based on division of a standard deviation of equivalent circle diameters of 100 of the plurality of laser irradiation marks by an average value is 3% or more, wherein a number of the plurality of laser irradiation marks is 100.

7. A method for manufacturing an amorphous metal ribbon, comprising:

preparing a material ribbon having a main surface made of an amorphous metal; and

applying laser processing to the main surface of the material ribbon to obtain the amorphous metal ribbon having a plurality of laser irradiation mark rows each including a plurality of laser irradiation marks arranged in a row, wherein

a distance between the plurality of laser irradiation mark rows that are adjacent to each other is set as d**1**,

a distance between the plurality of laser irradiation marks in each laser irradiation mark row is set as d**2**,

a diameter of a laser irradiation mark of the plurality of laser irradiation marks is set as d**3**,

a depth of the laser irradiation mark which is set as d**4** is 0.0055 mm or more and 0.15 mm or less, and

a volume occupancy rate V of the plurality of laser irradiation marks is set as (1/d**1**×d**3**×d**4**)×(1/d**2**)×100, the volume occupancy rate V of the plurality of laser irradiation marks is 0.00057% or more and 0.010% or less.

8. A magnetic core, comprising:

the amorphous metal ribbon according to claim **1**.

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